

Machine Learning: short introduction (trees and ANN), part 1

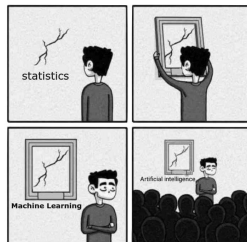
Sergey Korpachev on behalf of Dépôt

Outline

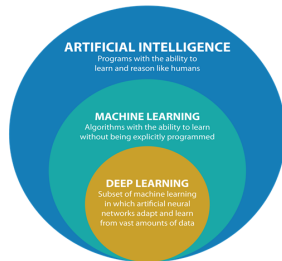
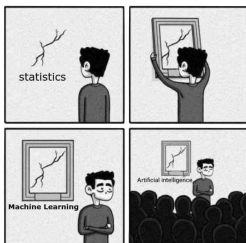
- Machine learning (ML)
 - Data
 - Features in ML
 - ML pipeline
 - Linear model
 - Decision tree
 - Neural network
 - Summary

Machine learning

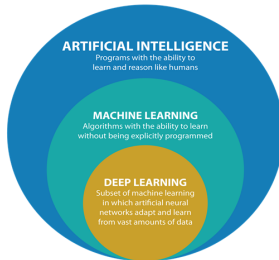
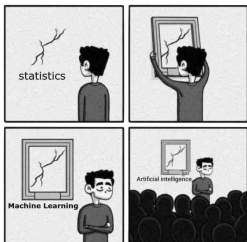
Machine learning —○ What is it?



Machine learning —○ What is it?

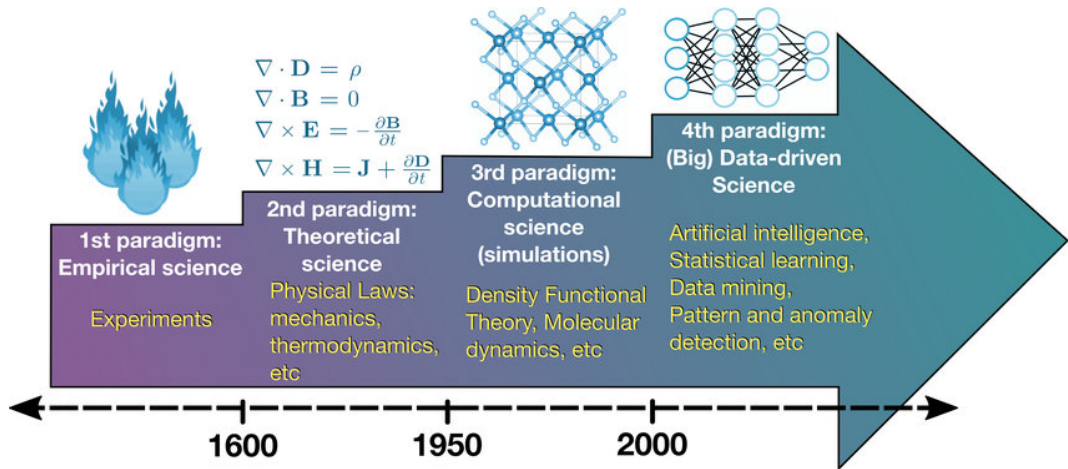


Machine learning —○ What is it?



Machine learning

—○ The Fourth Paradigm (Tony Hey, ...)



Data

Data

—○ Data?

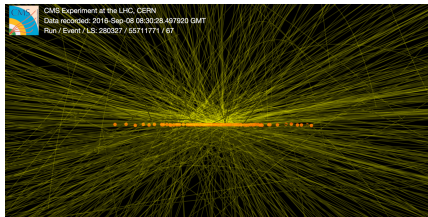
In the beginning was the Data:

- Information about goods and sales
- Data from subsystems of detector
- Subscribers database

Shall one question the Data:

- predict demand for this product?
- estimate energy of a particle?
- like or dislike the post?

How can we answer these questions?



Data

—○ Data?

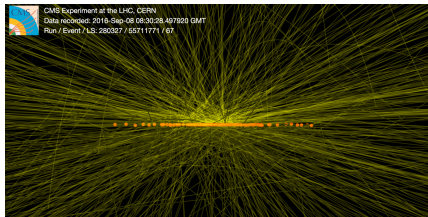
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How can we answer these questions? → **this lecture**

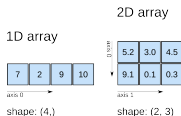


Data

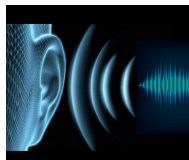
—○ What is data?

Anything can be data:

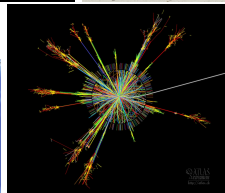
- 1 Numbers
- 2 Text
- 3 Images
- 4 Sound
- 5 Geomap
- 6 Particle collisions
- 7 Knowledge
- 8 You name it



3D array



18.
Shall I compare thee to a *Summer's day*?
Thou art more lovely and more temperate:
Rough winds do shake the darling buds of *May*,
And *Summer's* lease hath all too short a date:
Sometime too hot the eye of *Heaven* shines,
And often is his gold complexion dimm'd;
And every fair from *fair* *loose-time* declines,
By chance, or *nature's* changing course untrimm'd;
But thy eternal *Summer* shall not fade,
Nor lose possession of that *fairest* thou ow'st;
Nor shall death brag thou *wastest* it in his *shade*,
When in *eternal* lines to *time* thou grow'st;
So long as *men* can *breathe*, or *eyes* can *see*,
So long *lives* this, and *this* gives *life* to thee,



Features in ML

Features in ML

—○ Can we use data out of the box?

- Most likely, no: garbage in, garbage out!
- Main drawbacks of raw data:
 - Not in ML comprehensive format
 - Can have missing values or might be noisy
 - Useless/lack of useful features
- ! Analysing data improves its understanding



Features in ML

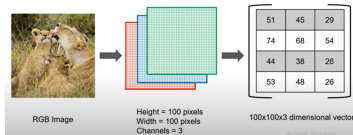
—○ Comprehensive format [more in the seminar!](#)

ML algorithms operate only with numerical sets, therefore:

- categorical data require one hot encoding (or any different method)
- text should also be vectorized
- pictures can be treated similarly

Color			
Red	Red	Yellow	Green
Red	1	0	0
Yellow	1	0	0
Green	0	1	0
Yellow	0	0	1

	good	movie	not	a	did	like
good movie	1	1	0	0	0	0
not a good movie	1	1	1	1	0	0
did not like	0	0	1	0	1	1



Features in ML

— Data preprocessing

more in the seminar!

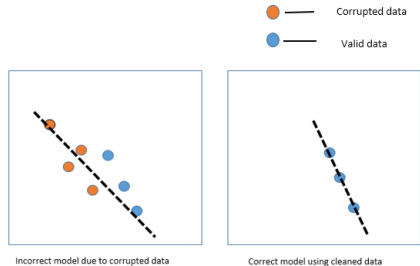
Main techniques:

- Imputation of missing data
- Data cleaning
- Handling outliers
- Shuffling
- Balancing classes
- Partitioning

#	Id	Name	Birthday	Gender	IsTeacher?	#Students	Country	City
1	111	John	31/12/1990	M	0	0	Ireland	Dublin
2	222	Mery	15/10/1978	F	1	15	Iceland	
3	333	Alice	19/04/2000	F	0	0	Spain	Madrid
4	444	Mark	01/11/1997	M	0	0	France	Paris
5	555	Alex	15/03/2000	A	1	23	Germany	Berlin
6	555	Peter	1983-12-01	M	1	10	Italy	Rome
7	777	Calvin	05/05/1995	M	0	0	Italy	Italy
8	888	Roxane	03/08/1948	F	0	0	Portugal	Lisbon
9	999	Anne	05/09/1992	F	0	5	Switzerland	Geneva
10	101010	Paul	14/11/1992	M	1	26	Italy	Rome

Annotations:

- Missing values: Row 2, City column.
- Invalid values: Row 5, Gender column (A); Row 6, Birthday column (1983-12-01).
- Misfielded values: Row 7, City column (Italy).
- Misspellings: Row 2, Name column (Mery); Row 8, Country column (Portugal).
- Uniqueness: Row 6, Id column (555).
- Formats: Row 6, Birthday column (1983-12-01).
- Attribute dependencies: Row 9, #Students column (5).



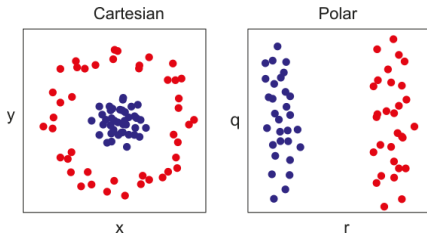
Features in ML

—○ Feature engineering

more in the seminar!

- **Construction:** vectorization techniques + your domain knowledge
- **Transformation:** functions, scaling, binning, cropping, resizing, blurring
- **Selection:** PCA, statistical tests, feature importance, hand-picking

Example: Using using Polar coordinates it's much easier to separate **red** dots from **blue** than using Cartesian coordinates:



Features in ML

—○ Not HEP

Supervised learning

Classification

- cat, dog or muffin
- relevant or spam
- disease or not
- good or bad
- ...

Regression

- rent price
- temperature
- annual profit
- driving time
- ...

Don't forget about **unsupervised learning**, **reinforcement learning**, **semi-supervised learning** and so on.

Supervised learning

Classification

- b , c , uds jet
- π , K , μ particle
- $t\bar{t}$ or QCD event
- select or reject trigger candidate
- ...

Regression

- energy resolution
- pile-up mitigation
- ...

Don't forget about **unsupervised learning**, **reinforcement learning**, **semi-supervised learning** and so on.

Features in ML

—○ ML challenges: slide 1 (A. Ustyuzhanin)

- Precise and fast particle tracking (single tracks, shower, jets)
- Particle identification
- Fast and accurate online data processing and filtering
- Anomaly detection (data quality monitoring, infrastructure monitoring)
- Detector design optimization (bayesian optimization, surrogate modelling)
- Data analysis (signal from background separation, ...)
- Simulation (speed-up simulation using generative models, simulator parameters optimization - tuning)
- ...

Features in ML

—○ ML challenges: slide 2 (A. Ustyuzhanin)

Tracking system features

- Particle momentum
- Particle charge
- Track parameters
- Quality of track fit
- Number of track hits
- ...

RICH features

- Angle θ
- Quality of angle reconstruction
- Reconstructed particle type
- Reconstructed particle energy
- Light intensity
- ...

Calorimeter features

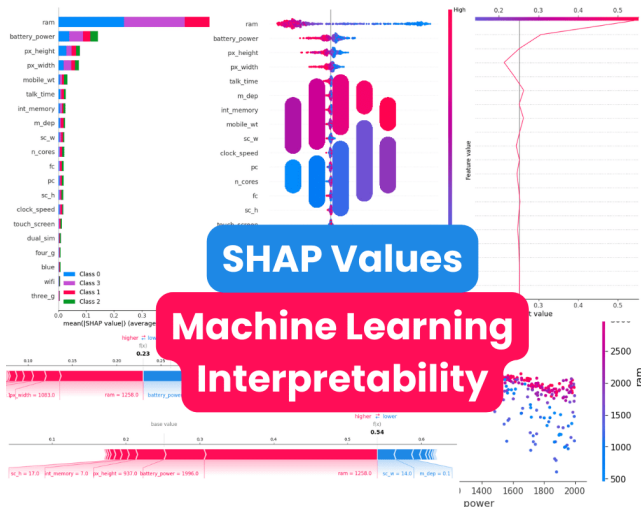
- Measured particle energy
- Shower parameters: length, width, ...
- Number of clusters in each layer Intensity of the clusters
- Intensity of the clusters
- Distance form track of the original particle
- ...

Muon detector features

- Muon track parameters
- Quality of track fit
- Number of active layers
- Distance between the track and the active layers
- Length of shower
- ...

Features in ML

—○ SHAP value

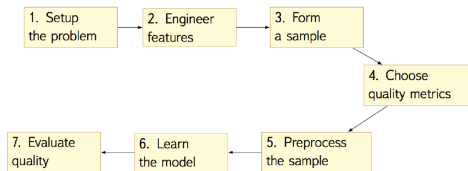


ML pipeline

ML pipeline

—○ What is ML pipeline?

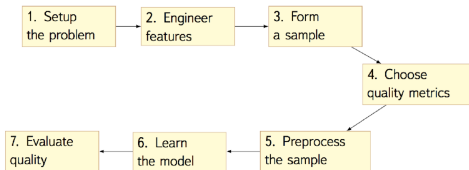
Machine Learning Pipeline



ML pipeline

—○ What is ML pipeline?

Machine Learning Pipeline



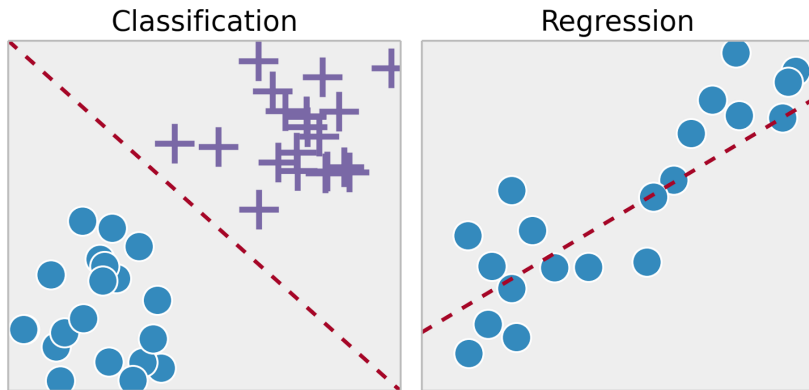
Machine Learning Pipeline



Linear model

Linear model

—○ Classification and regression



Linear model

—○ Regression

- $Y = \mathbb{R}$
- N objects with K real features: $D = \mathbb{R}^K$
- $a(\mathbf{x}, \boldsymbol{\theta}) = \theta_0 + \sum_{j=1}^K f_j(\mathbf{x}) \cdot \theta_j$
- Extend and reassign:
 $[1, f_1(\mathbf{x}), \dots, f_K(\mathbf{x})] \equiv \mathbf{x}$
 $[\theta_0, \dots, \theta_K] \equiv \boldsymbol{\theta}$
- Then $a(\mathbf{x}, \boldsymbol{\theta}) = \langle \mathbf{x}, \boldsymbol{\theta} \rangle$
- Minimization problem:
 - $\mathcal{L}(\boldsymbol{\theta}) = (\langle \mathbf{x}, \boldsymbol{\theta} \rangle - y)^2$
 - $Q(\boldsymbol{\theta}) = \frac{1}{N} \sum_{i=1}^N (\langle \mathbf{x}_i, \boldsymbol{\theta} \rangle - y_i)^2$
- Then we need to minimize $Q(\boldsymbol{\theta})$ by varying $\boldsymbol{\theta}$:
 - $\hat{\mathbf{a}} = \arg \min_{\boldsymbol{\theta} \in \Theta} Q(\boldsymbol{\theta})$

- $Y = \{-1, +1\}$
- N objects with K real features: $D = \mathbb{R}^K$
- $a(x, \theta) = \theta_0 + \sum_{j=1}^K f_j(x) \cdot \theta_j$
- Extend and reassign:
 $[1, f_1(x), \dots, f_K(x)] \equiv [1, x_1, \dots, x_K] \equiv x$
 $[\theta_0, \dots, \theta_K] \equiv \theta$
- Then $a(x, \theta) = \langle x, \theta \rangle$
- Minimization problem:
 - $\mathcal{L}(\theta) = [\text{sign}\langle x, \theta \rangle \neq y]$
 - $Q(\theta) = \frac{1}{N} \sum_{i=1}^N [\text{sign}\langle x_i, \theta \rangle \neq y_i]$
- Then we need to minimize $Q(\theta)$ by varying θ :
 - $\hat{a} = \arg \min_{\theta \in \Theta} Q(\theta)$

Linear model

—○ Solution can be found analytically

After defining feature matrix as:

$$F = \begin{bmatrix} f_1(\mathbf{x}_1) & \cdots & f_K(\mathbf{x}_1) \\ \vdots & \ddots & \vdots \\ f_1(\mathbf{x}_N) & \cdots & f_K(\mathbf{x}_N) \end{bmatrix}$$

Minimization problem for linear regression:

- $Q(\boldsymbol{\theta}) = \frac{1}{N}(F\boldsymbol{\theta} - \mathbf{y})^T(F\boldsymbol{\theta} - \mathbf{y}) = \boldsymbol{\theta}^T F^T F \boldsymbol{\theta} - 2\mathbf{y}^T F \boldsymbol{\theta} + \mathbf{y}^T \mathbf{y}$
- $\frac{\partial Q}{\partial \boldsymbol{\theta}^T} = 2F^T F \boldsymbol{\theta} - 2F^T \mathbf{y} = 0$

$$\hat{\boldsymbol{\theta}} = (F^T F)^{-1} F^T \mathbf{y}$$

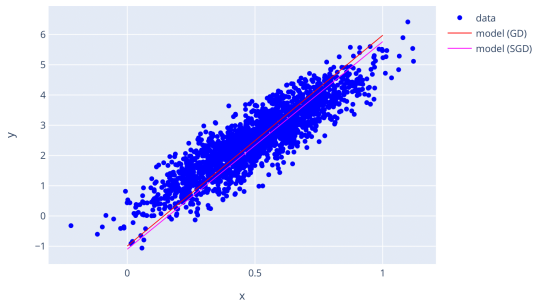
Problem 1: calculation of $(F^T F)^{-1}$ has **complexity of $O(K^5 N)$** $\Rightarrow$ computationally-intensive for large number of features

Problem 2: F is often **ill-conditioned** $\Rightarrow$ computational errors causes instabilities and extremely large solutions

Linear model

—○ Example

- Let's consider a problem where objects have only one feature:
 $\mathbf{x} = [1, f_1(\mathbf{x})]$
- Then $\boldsymbol{\theta} = [\theta_0, \theta_1]$
- How can we find minimum of $Q(\boldsymbol{\theta})$ without solving matrix equation?

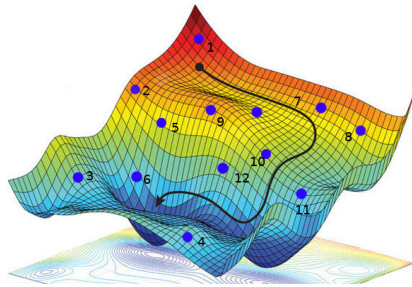


Linear model

—○ Random search

- Start with some vector θ^0
- Initialize $\hat{\theta} = \theta^0$
- ↓ Take randomly another vector θ'
- ↓ Compare $Q(\theta')$ and $Q(\hat{\theta})$
- ↓ $\hat{\theta} = \arg \min_{\hat{\theta}, \theta'} (Q(\hat{\theta}), Q(\theta'))$
- ↓ Repeat until convergence

Problem: random search of $\hat{\theta}$ is not effective for practical use in high dimensional Θ space



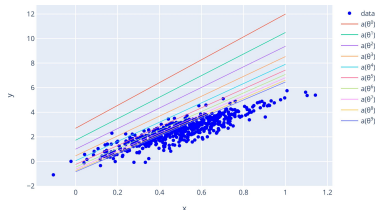
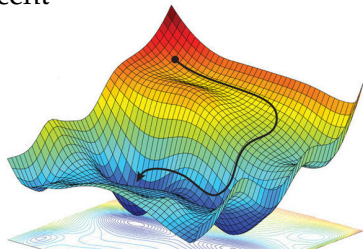
Q as a function of θ_0 and θ_1 .
Blue dots represent Q values for different θ

Linear model

—○ Gradient descent

Idea: bring up additional information about the direction to the minimum of Q using its **antigradient**

- Start with some vector θ^0
- $\hat{\theta} \leftarrow \theta^0$
- ↓ Calculate direction to the minimum:
$$-\nabla Q(\hat{\theta}) = -\left[\frac{\partial Q}{\partial \theta_0}, \frac{\partial Q}{\partial \theta_1}\right]_{\hat{\theta}}$$
- ↓ Move towards the minimum:
$$\hat{\theta} \leftarrow \hat{\theta} - \eta \nabla Q(\hat{\theta})$$
- ↓ Repeat until convergence



Section summary

- All ML problems start from:
 - Data preprocessing
 - Feature engineering
- Type of model/approach is defined by our intentions:
 - **SL**: seek for a **predictive model**: for a given object based on its **features** returns the **answer**
 - classification
 - regression
 - ranking
 - **UL**: look for insights into the **structure of data**
 - **RL**: want to build an agent capable of **acting in the given environment**
- Model is constructed through the **training process** involving **loss function optimization**
- Once we are sure that the model is reasonable we can **deploy it in production**
- **Linear models** are the simplest example of models used for **SL**
- Solution: analytically, but **(stochastic) gradient descent** is a more general approach